Advanced Machine Learning (AML)

Traffic flow prediction and adaptive signal control

Spatio-temporal forecasting via graph attention networks and flow models

Team Quanta(T020)

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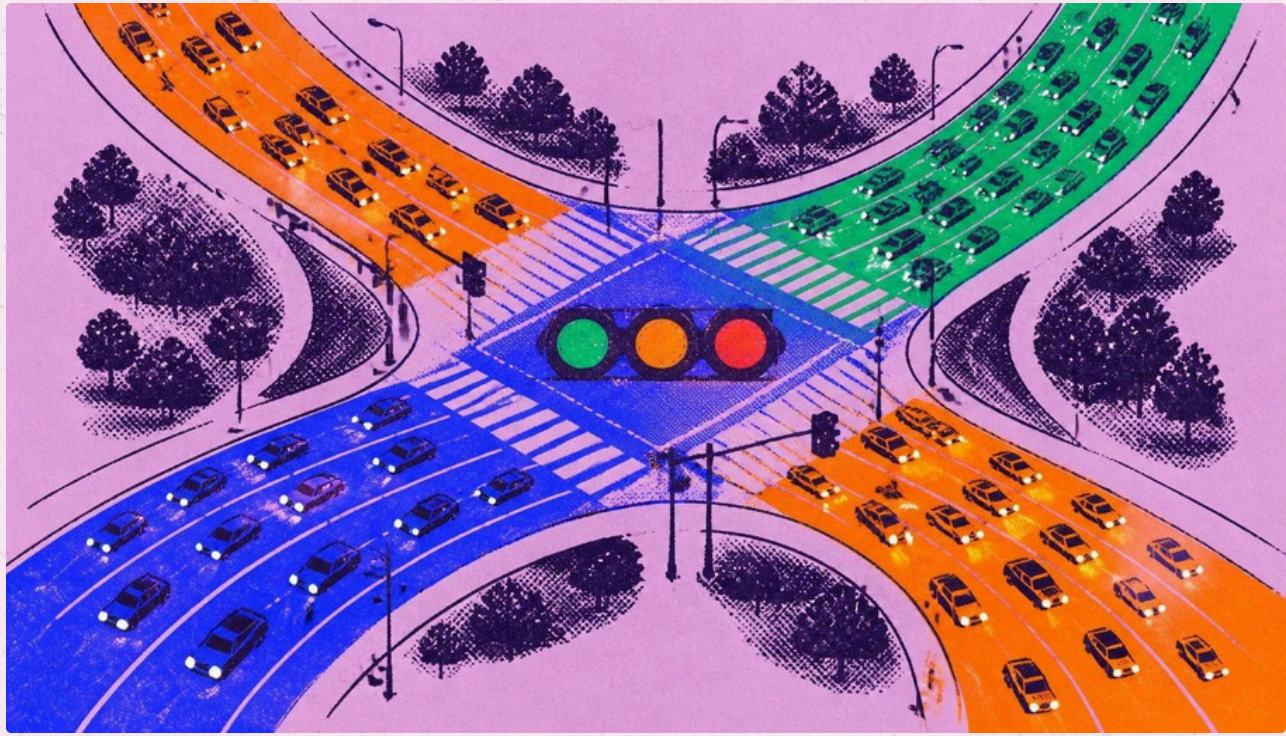
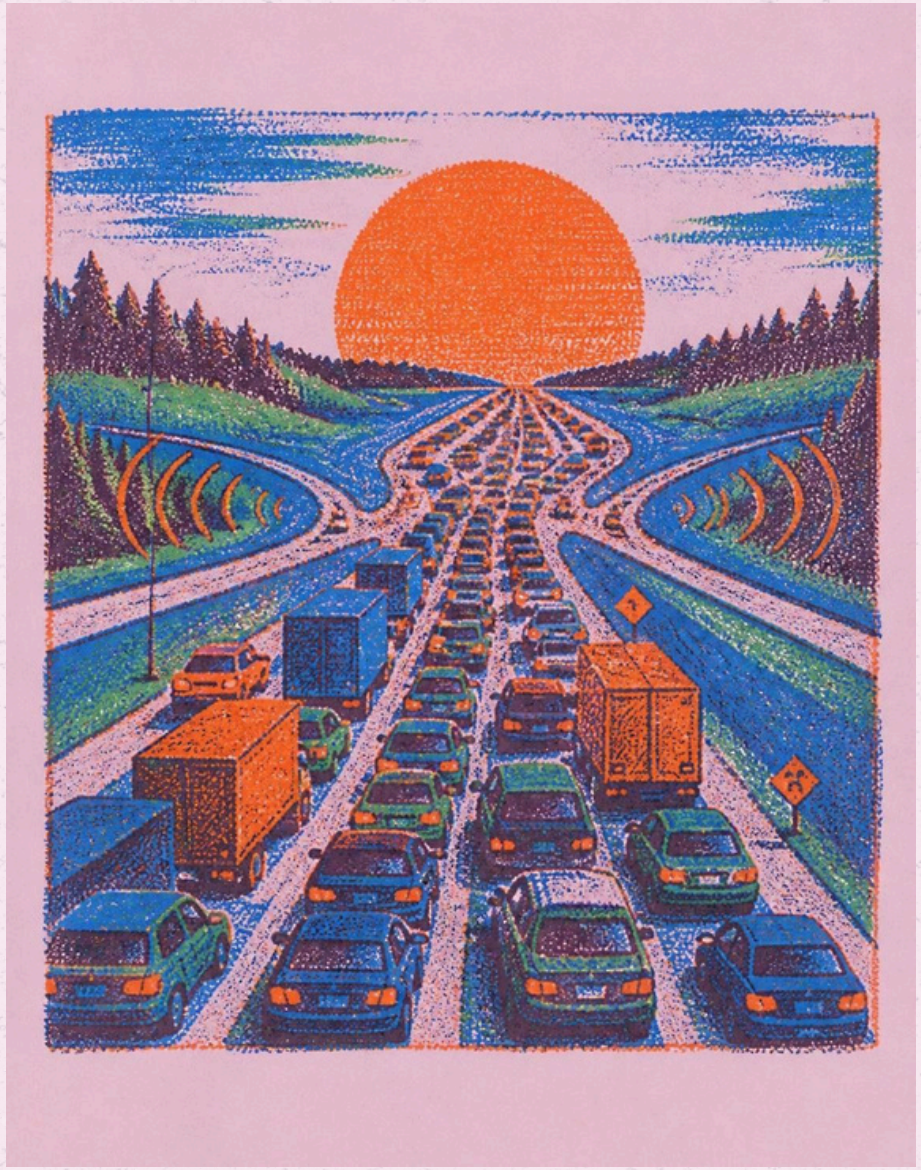


GitHub Repository & Reproducibility Suite



Problem statement

Isolated traffic forecasting is not enough

Trafficvolatility:highway speedsswingnon-linearly with rush-hour congestion and abrupt shockwaves. **Spatial cascades:** a bottleneck at one merge ripples backwards across upstream segments.

Research question: can deep Graph Attention Networks (GAT) beat tuned tabular ML on multi-horizon speed forecasting? **Downstream goal:** turn speed forecasts into adaptive signal timing that reduces delay.

Methodology

A five-stage machine learning pipeline

From 34k loop detector timestamps to macroscopic flow-controlled traffic intersections.

1

Stage 01

Data ingestion

Audit 34,272 timestamps across 207 Los Angeles loop detectors.

2

Stage 02

Feature extraction

Causal speed lags (t, t-5m, ..., t-60m) and cyclic time encodings.

3

Stage 03

Model comparison

Persistence, multi-output Random Forest, pooled XGBoost, and GAT.

4

Stage 04

Multi-horizon testing

Quantify error growth at 5, 15, 30 and 60 minutes out.

5

Stage 05

Signal simulation

Couple forecasts with Greenshields flow theory to test adaptive control.

Dataset audit

Two highway networks, one rush-hour pattern

METR-LA (primary)

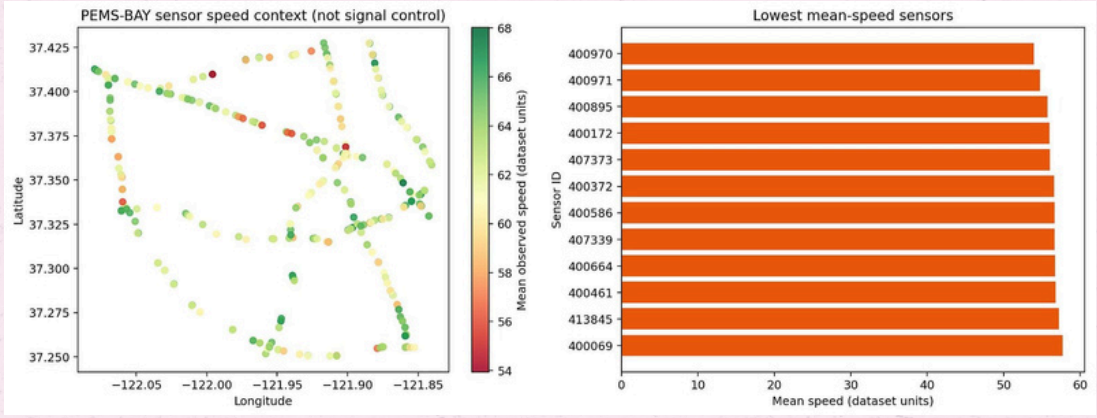
207 sensors, 34,272 timestamps at 5-min intervals, 7.09M observations, 8.11% missing.

PEMS-BAY (validation)

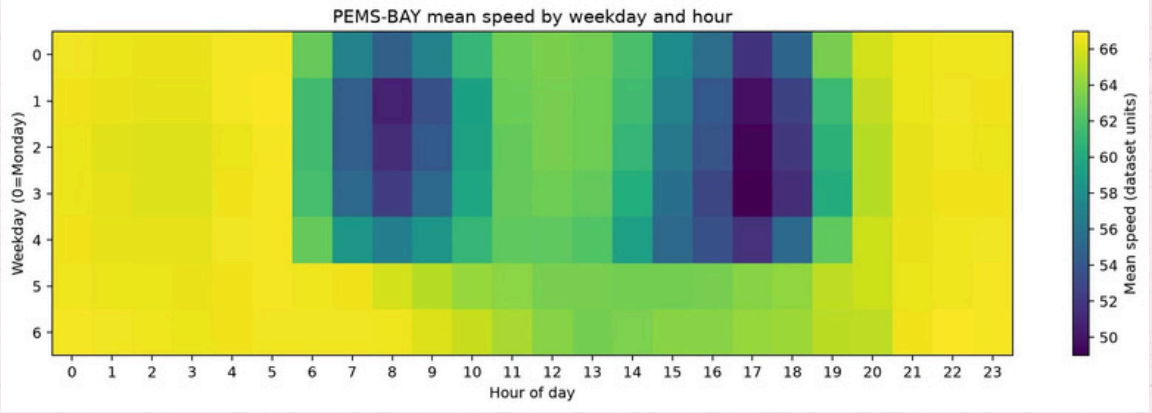
325 sensors, 52,116 timestamps, 16.9M observations, 0.003% missing.

Diurnal pattern

Sharp weekday speed dips at 07:00–09:00 and 16:30–18:30; weekends flow freely.



PEMS-BAY spatial sensor coverage and bottleneck stations with lowest mean velocity.



Recurring weekday rush-hour dips (08:00 & 17:00) vs. free-flowing weekend traffic.

Representation engineering

Causal spatio-temporal predictors

Speed lags

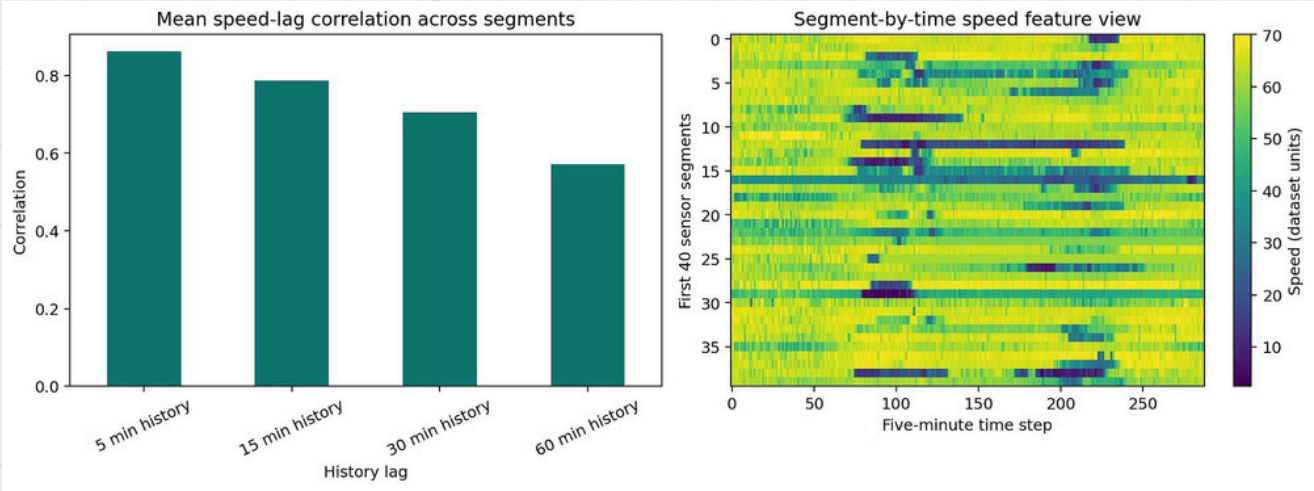
Speeds at t, t-5m, t-10m, t-15m, t-30m, t-60m capture short-term inertia and long-term trend.

Cyclic time encodings

Sine and cosine of minute-of-day (1440) and day-of-week (7) prevent midnight boundary discontinuity.

Road graph

Sensor distances through a thresholded Gaussian kernel produce 1,313 directed topological edges.



Immediate 5–30 min speed lags provide over 85% of predictive power.

Model architecture

Spatio-Temporal Forecasting Architectures

Persistence

Future speed equals current speed.

Segment-wise Random Forest

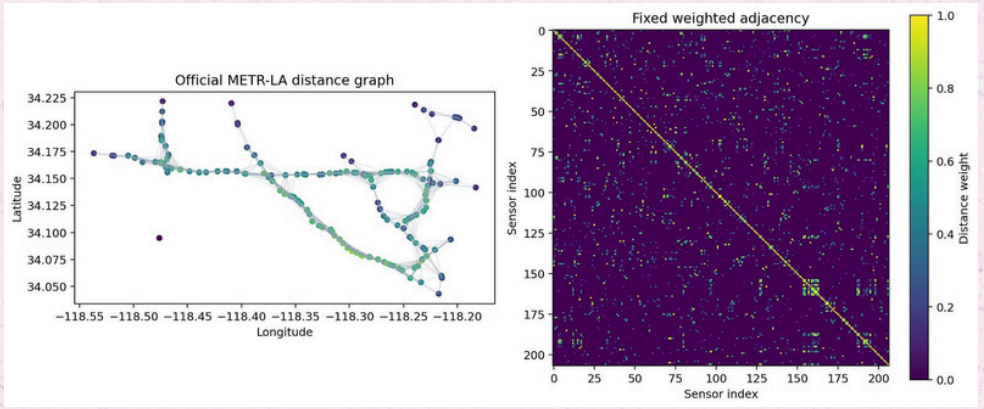
207 multi-output regressors predicting 5, 15, 30 and 60 min.

Pooled XGBoost

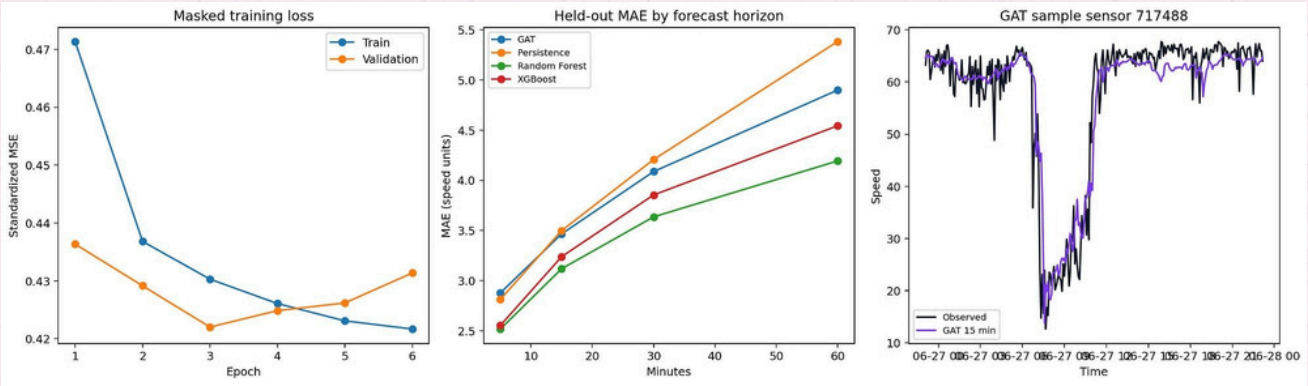
One boosted-tree model with segment ID encodings.

GAT

Attention weights across 1,313 edges aggregate upstream and downstream neighbours.



METR-LA topology of 207 sensors linked by roadway network distance.



Masked loss convergence stopping at Epoch 3.

Evaluation protocol

Leakage-free chronological backtesting

Train (70%)

Mar 1 – Apr 30, 2012

23,990 timestamps. Used for fitting tree ensembles and learning GAT graph attention parameters.

Validation (10%)

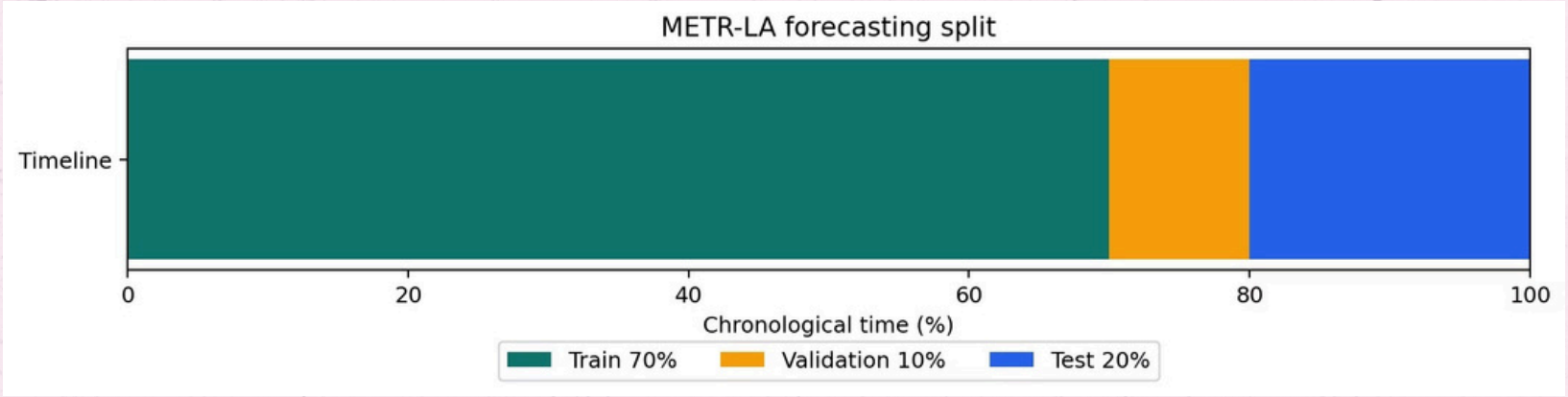
May 1 – May 14, 2012

3,427 timestamps. Dedicated exclusively to early stopping, tree depth tuning, and hyperparameter selection.

Test (20%)

May 15 – Jun 27, 2012

6,855 timestamps, yielding 1,244,846 evaluation values per horizon. Scored via MAE (mph), RMSE, and MAPE (%).



Strict 70/10/20 chronological split eliminating temporal lookahead bias.

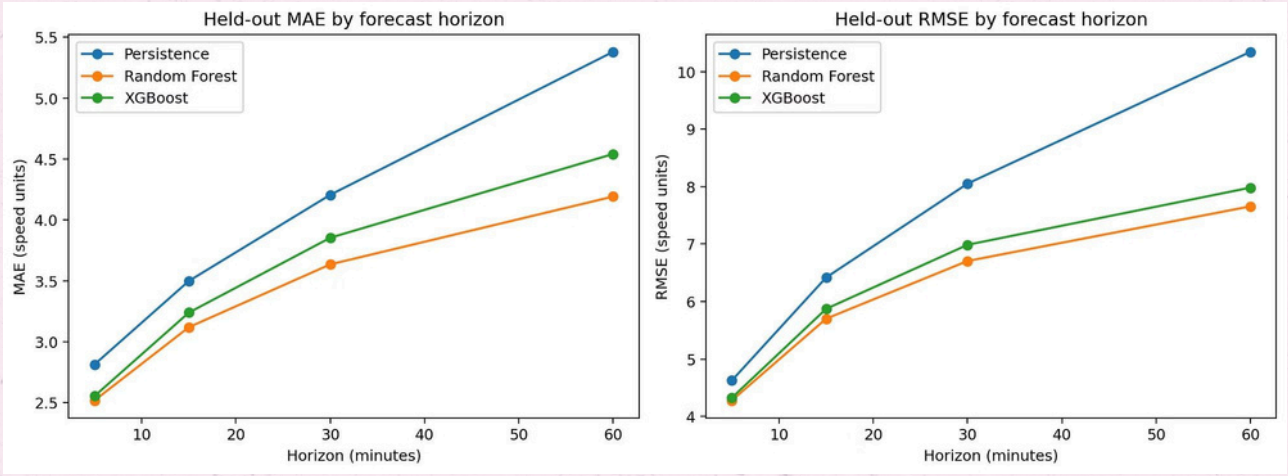
Benchmark results

Random Forest wins at every horizon

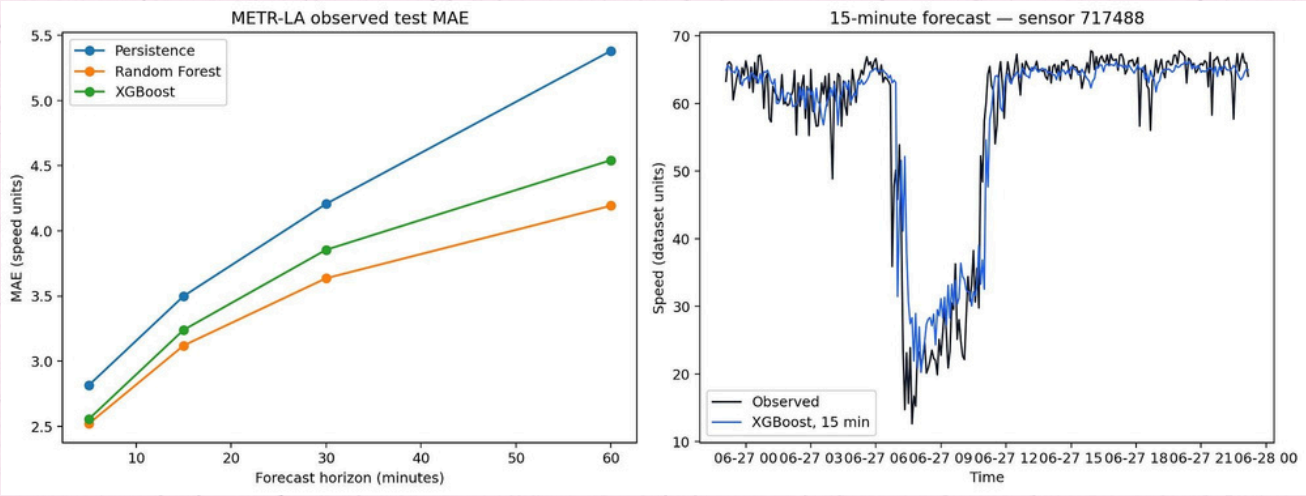
TestMAE inmph (lower is better).Random Forest cut 60-min MAE by **22.1%** versus Persistence.

Horizon	Persistence	GAT	XGBoost	Random Forest	RF advantage
5 min	2.814	2.879	2.554	**2.519**	-10.5%
15 min	3.499	3.466	3.239	**3.119**	-10.9%
30 min	4.207	4.087	3.854	**3.635**	-13.6%
60 min	5.381	4.898	4.542	**4.192**	-22.1%

RF advantage calculated relative to Persistence baseline. Segment-wise models beat pooled architectures.



Macro MAE & RMSE scaling across 5m–60m.

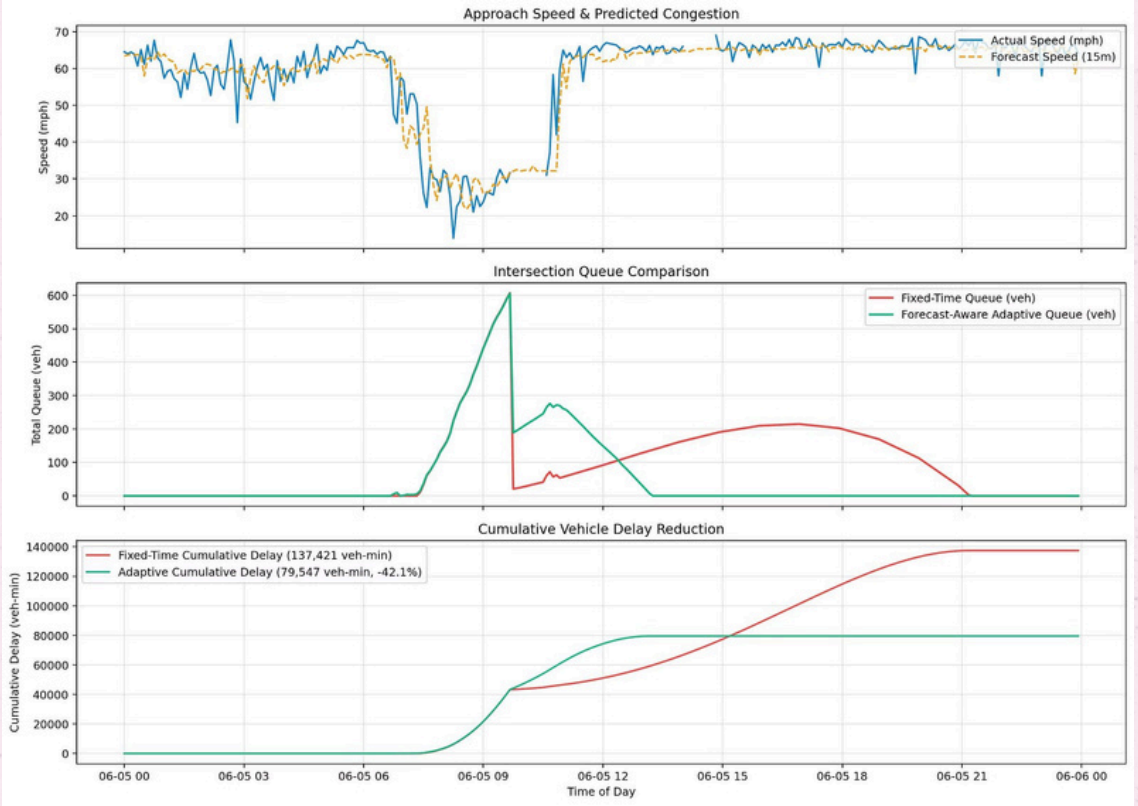


XGBoost 15-min forecast tracking real rush-hour plunge.

Adaptive signal simulation

Forecast-aware signals cut simulated delay by 42.11%

- **Peak volatility (RF):** off-peak MAE 2.81 to 3.64 mph; rush-hour MAE 4.34 to 6.37 mph, about 75% higher
- **Simulation:** forecast speeds coupled with Greenshields flux (q = k·v) adjust green times
- **Delay:** fixed-time 137,420.9 veh-min vs adaptive 79,546.6 veh-min
- **Impact:** 42.11% less cumulative vehicle delay (macroscopic numerical simulation)



Forecast-adaptive green time cuts cumulative vehicle delay by 42.11%.

Conclusions & future directions

Takeaways, limits and next steps

Takeaways

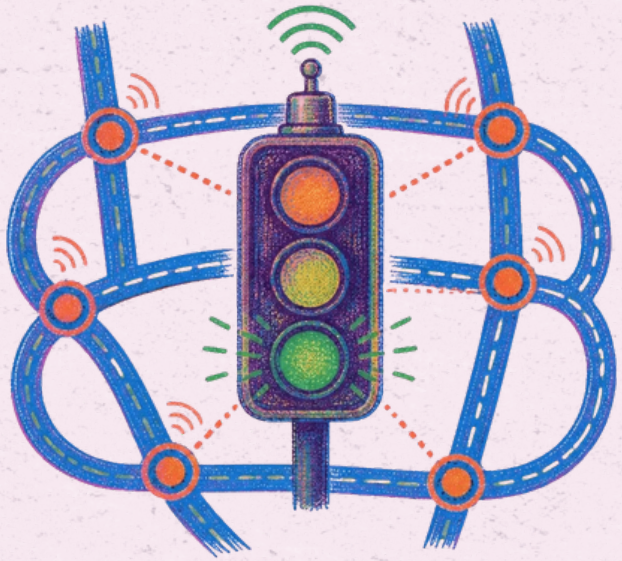
Segment-wise Random Forest was most accurate; adaptive signals gave a 42.11% simulated delay reduction.

Limitations

Standard GAT trailed lag-feature models without temporal convolutions; simulation assumes macroscopic flow.

Future work

Spatio-temporal GNNs (STGCN, DCRNN) and microscopic multi-agent validation in SUMO.



Thank You !!

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